Iteration 1 Report

e-Project

Team 6

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Overview

During this iteration, we implemented the basic authentication functionality of the e-Project web application according to the requirements of Sprint 1. The main goal of this sprint was to allow users to register, log in, and log out of the system, as well as display a welcome screen when visiting the home page.

The project was implemented using ASP.NET Core MVC with C# and a relational database to store user accounts. We created the necessary controllers, models, views, and database entities required for user authentication.

The system now allows anonymous users to register through a “Register” link visible in the top menu on all pages, and log in via a “Login” link in the top menu also visible on all pages. Logged-in users can log out via the “Logout” link in the top menu visible on all pages, which redirects them to the home page of the application. The navigation bar dynamically changes depending on whether the user is authenticated or not. Proper validation is performed by the system, and detailed error messages are displayed.

Individual Contributions

Arina:

Arina implemented the login functionality of the application in the AccountController, which verifies the email and password entered by the user. The controller checks if the email exists in the database and then compares the password with the stored value. If the credentials are correct, the user is authenticated using cookie authentication and redirected to the home page.

Arina also created the LoginViewModel, which is responsible for validating the login form inputs, such as email and password.

Additionally, she implemented the navigation bar and home screen functionality. The navigation bar changes depending on whether the user is logged in or not. Anonymous users see Login and Register links, while authenticated users see a Logout option. The home page also displays the required welcome message: “Welcome to e-Project”.

Arina also created the UserAccount entity, which defines the structure of the user table in the database. The entity includes fields for email, password, name, and phone number, along with validation rules.

Example code fragment for login validation:

```
var user = await _context.UserAccounts
   .FirstOrDefaultAsync(u => u.Email.ToLower() == model.Email.ToLower());

if (user == null)
{
   ModelState.AddModelError("", "Email does not exist in the system.");
}
```

Magdalena:

Magdalena set up the project repository on GitHub and initialized the project structure. She also configured the database connection and ensured that the application could store user data in the database.

She implemented the user registration functionality. This includes creating the RegisterViewModel and adding the registration logic to the AccountController. The registration process validates the user's inputs and ensures that the email address is unique before a new account is created.

Example code fragment for checking unique email addresses:

```
bool emailExists = await _context.UserAccounts

  .AnyAsync(u => u.Email.ToLower() == model.Email.ToLower());



if (emailExists)

{

  ModelState.AddModelError(nameof(model.Email), "This email is already registered.");

}
```

Magdalena also implemented the Register and Login views and handled the styling of the authentication pages. The forms were styled using Bootstrap and placed inside centered cards to create a cleaner interface.

She also added success messages for authentication actions, which notify users when they successfully register, log in, or log out.

Additionally, Magdalena implemented password hashing for improved security. Instead of storing passwords in plain text, the application uses ASP.NET's PasswordHasher to

hash user passwords before saving them to the database. During login, the entered password is verified against the stored hash, ensuring that user credentials are stored securely.

Id	Email	Password	Name	Phone
5	test@example.c...	AQAAAAIAAYa...	Test User	0886259333
6	magdalenazhel...	AQAAAAIAAYa...	Magdalena Zhe...	0886259335
NULL	NULL	NULL	NULL	NULL

Acceptance Tests

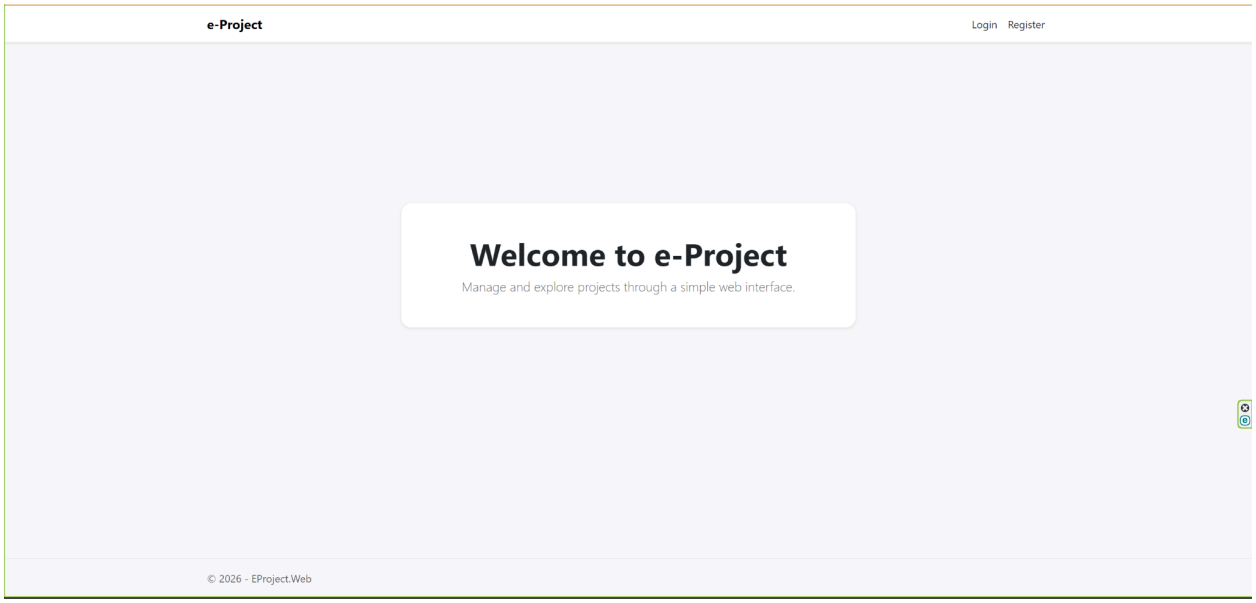
Test 1 – Welcome Screen

User Story: As a user, I want to see a welcome screen when visiting the home page.

Input: User navigates to the main URL of the application.

Expected Output: The page displays the message: Welcome to e-Project

Result: The home page correctly displays the welcome message for both anonymous and logged-in users.



Test 2 – User Registration

User Story: As an anonymous user, I want to register in the e-Project system.

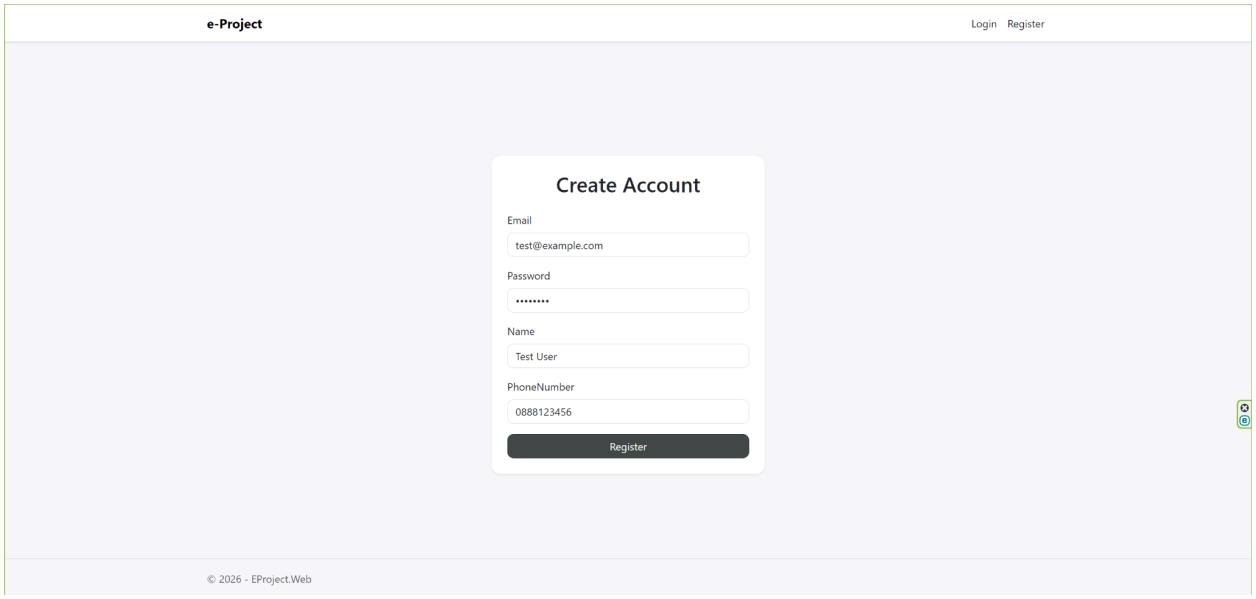
Input:
Email: test@example.com
Password: abc12345
Name: Test User
Phone: 0888123456

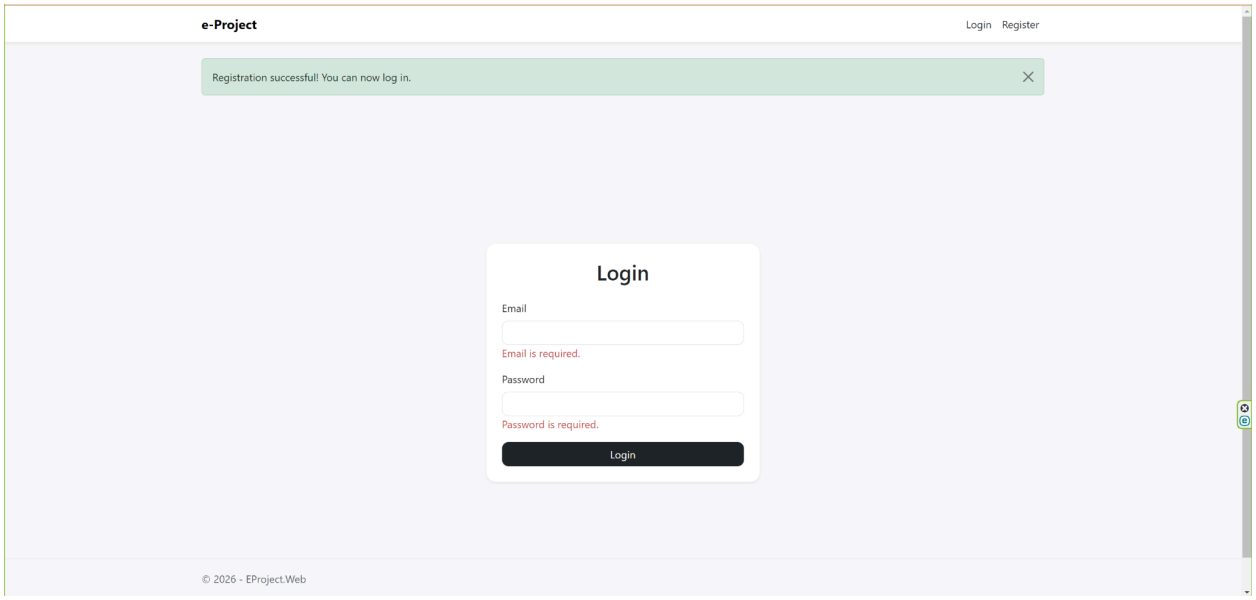
Expected Output

- The system validates the input fields
- The user account is created
- The data is stored in the database
- The user is redirected to the login page

Result

The registration works successfully and the new user appears in the database.





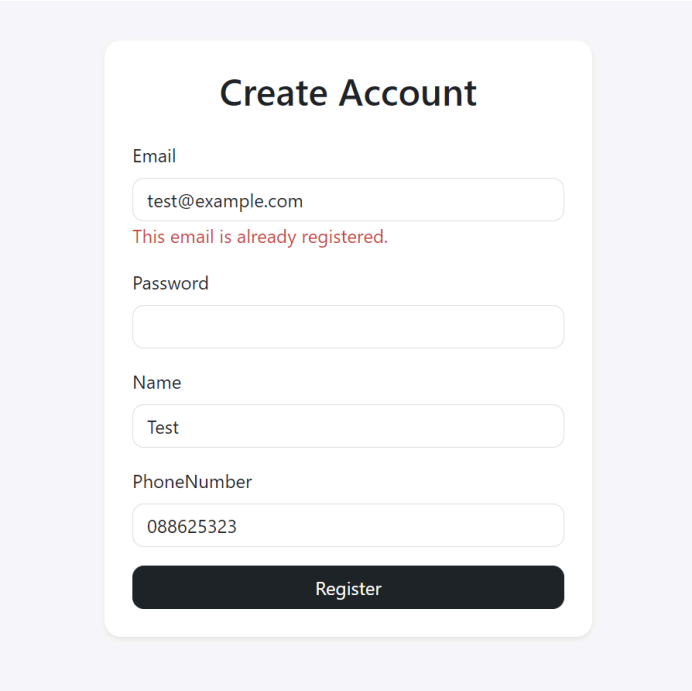
Test 3 – Duplicate Email Validation

User Story: The system should prevent two users from registering with the same email.

Input: Attempt to register with an email that already exists in the database.

Expected Output: This email is already registered.

Result: The system correctly prevents duplicate email registrations.



Test 4 – User Login

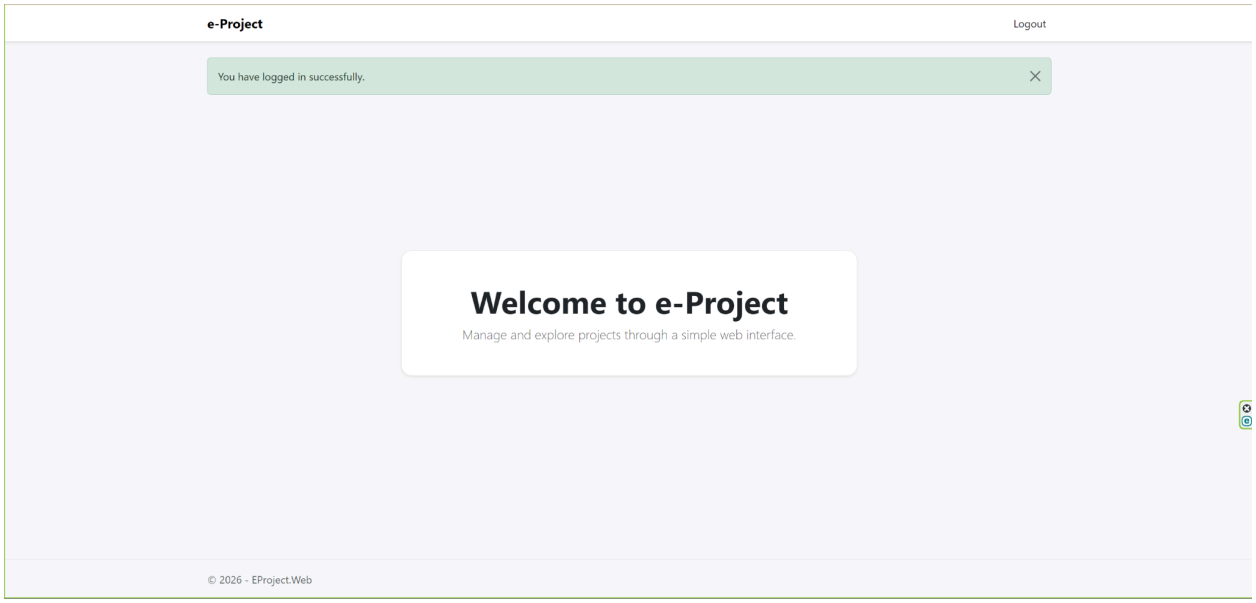
User Story: As a registered user, I want to log in using my email and password.

Input
Email: test@example.com
Password: abc12345

Expected Output

- User is authenticated
- User is redirected to the home page
- Navigation bar changes to show Logout

Result: The login functionality works correctly and the navigation updates accordingly.

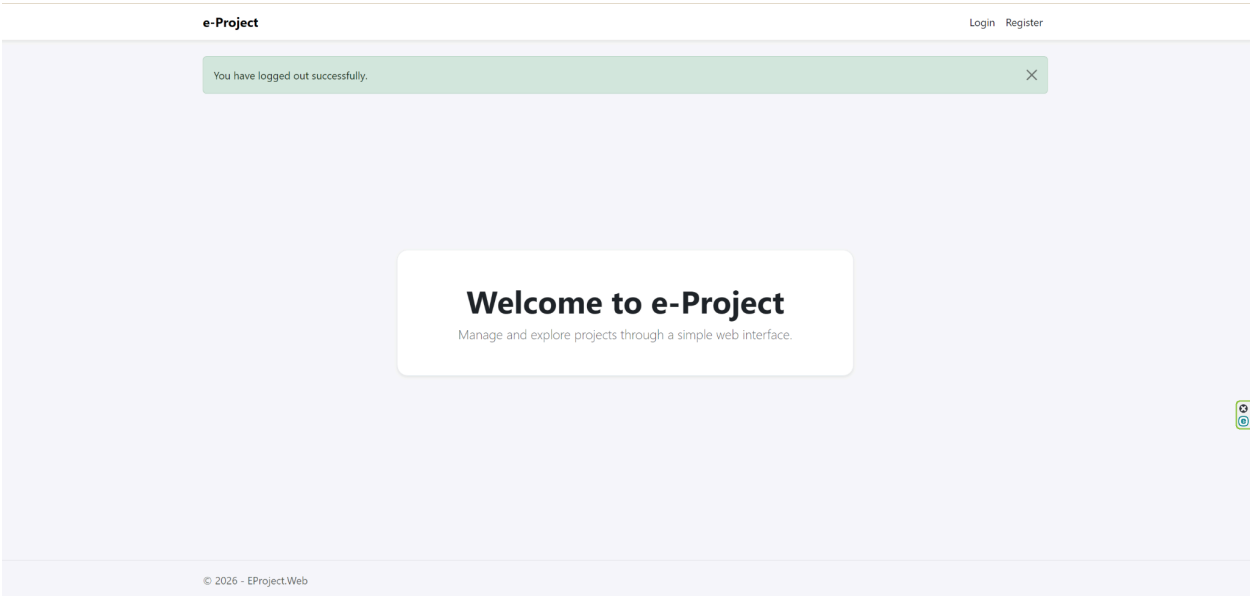


Test 5 – Logout

User Story: As a logged-in user, I want to log out from the system.

Input: User clicks the Logout button in the navigation bar.

Expected Output:

- The user is logged out
- The user is redirected to the home page
- Navigation bar displays Login and Register again

Result: The logout functionality works correctly.